

Intermediate-Elective

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Set Up

Packages

```
# general use and cleaning

library(tidyverse)
library(here)
library(janitor)
library(NatParksPalettes)

# reading in .xlsx files
library(readxl)
```

Data

```
birds <- read_csv(here("data", "birds.csv"))
```

Rows: 15377 Columns: 6

-- Column specification -----

Delimiter: ","

chr (4): season, species, repeat_observation, e_bird_group

dbl (1): count

date (1): observation_date

- i Use ``spec()`` to retrieve the full column specification for this data.
- i Specify the column types or set ``show_col_types = FALSE`` to quiet this message.

```
aq_ins <- read_xlsx(  
  here("data", "Aquatic Sampling Data-2026-03-10.xlsx"),  
  sheet = "Aquatic Insects")  
  
sites <- read_xlsx(  
  here("data", "Aquatic Sampling Data-2026-03-10.xlsx"),  
  sheet = "sites"  
)
```

Cleaning sites

```
# creating new object from sites  
ncos_sites <- sites |>  
  #filtering to only include NCOS quarterly sites  
  filter(sector == "NCOS" &  
    sampling_frequency == "Quarterly")
```

Cleaning aq_ins

```
# create a new object  
aq_ins_clean <- aq_ins |>  
  # clean column names  
  clean_names() |>  
  semi_join(ncos_sites, by = "site") |>  
  mutate(site_full = case_when(  
    site == "NVBR" ~ "North of Venoco Bridge",  
    site == "NEC" ~ "East Channel",  
    site == "NMC" ~ "Main Channel",  
    site == "NPB" ~ "South of Phelps Bridge",  
    site == "NPB1" ~ "North of Phelps Bridge",  
    site == "NPB2" ~ "Phelps Road",  
    site == "NWP" ~ "West Pond",  
    site == "NDC" ~ "Devereux Creek"))
```

TidyTuesday 03/11/2020 - IKEA Furniture

```
# Wrangle data to count entries per person and site
dt <- aq_ins_clean |>
  # remove rows where either person or site is missing
  filter(!is.na(person_entering_data), !is.na(site_full)) |>
  # group by every unique combination of site and person
  group_by(site_full, person_entering_data) |>
  # count the number of rows in each group, creating column "n"
  tally() |>
  # ungroup
  ungroup() |>
  # keep only person/site combinations with 2 or more entries, too crowded
  filter(n >= 2) |>
  # sort rows from highest to lowest count
  arrange(desc(n)) |>
  # convert site and person columns to factors
  mutate(site_full = as.factor(site_full),
         person_entering_data = as.factor(person_entering_data))

# Set number of empty bars between groups
# defines how many blank bars to insert as visual padding in the circular plot
empty_bar <- 2

# create a matrix of NA values with the same number of columns as dt
to_add <- matrix(NA, empty_bar, ncol(dt))
# assign column names to match dt so they can be row-bound together
colnames(to_add) <- colnames(dt)
# empty rows go to the bottom of dt
dt <- rbind(dt, to_add)
# assign a numeric ID to each row
# this ID becomes the x-axis position in the circular plot
dt$id <- seq(1, nrow(dt))

# calculate label angles
# copy dt so label angle calculations don't affect the plotting data
label_data <- dt
# total number of bars including empty ones
number_of_bar <- nrow(label_data)
# calculate the angle for each bar's label based on its position in the circle
angle <- 90 - 360 * (label_data$id - 0.5) / number_of_bar
```

```

# if the angle puts the label on the left side of the circle, right-align it
# otherwise left-align - keeps labels readable on both sides
label_data$hjust <- ifelse(angle < -90, 1, 0)
# flip labels on the left side of the circle 180 degrees so they don't appear upside down
label_data$angle <- ifelse(angle < -90, angle + 180, angle)

# Plot
p <- ggplot(dt) +
  # draw bars with position on x (by ID), height = count, colored by site
  geom_bar(aes(x = as.factor(id), y = n, fill = site_full), stat = "identity") +
  # set y-axis limits: negative lower bound creates the hollow center of the wheel;
  # upper bound gives room above the tallest bar
  ylim(-10, max(dt$n, na.rm = TRUE) + 10) +
  theme_minimal() +
  # apply custom color palette
  scale_fill_manual(values = natparks.pals("DeathValley")) +
  theme(
    # remove all axis tick labels
    axis.text = element_blank(),
    # remove axis titles
    axis.title = element_blank(),
    # remove background grid lines
    panel.grid = element_blank(),
    # set negative margins to maximize plot area in the circular layout
    plot.margin = unit(rep(-1, 4), "cm")
  ) +
  # add a centered title in the middle of the wheel
  annotate(geom = "text",
    x = 0, y = -10,          # x = 0 centers it; y = -10 places it in the hollow
    hjust = .5, vjust = 1,  # center-align horizontally, top-align vertically
    label = "NCOS Aquatic Invertebrate\nSamples by Data Enterer",
    size = 2.5, lineheight = .8,
    color = "black") +
  # wrap the bar chart into a circle
  coord_polar(start = 0) +
  # add person name labels at the end of each bar, rotated to follow the circle
  geom_text(data = label_data,
    aes(x = id, y = n + 1,          # place label just above the bar tip
        label = person_entering_data, # text to display
        hjust = hjust),             # use the calculated alignment
    color = "black", fontface = "bold", alpha = 0.6, size = 3,
    angle = label_data$angle,       # rotate label to match bar angle
  )

```

```
inherit.aes = FALSE)
```

p

Warning: Removed 2 rows containing missing values or values outside the scale range (`geom_bar()`).

Warning: Removed 2 rows containing missing values or values outside the scale range (`geom_text()`).

